ARUN R

Agricultural Engineering | Farm Mechanization | AI in Agriculture | CAD Designer | Smart Irrigation Systems

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PROFESSIONAL SUMMARY

Final-year B.Tech Agricultural Engineering student specializing in Farm Mechanization, Precision Agriculture, AI Automation, and Smart Irrigation Systems. Experienced in designing engineering solutions using AutoCAD and SolidWorks, with hands-on exposure to hydroponics, agricultural machinery, and AI-powered farming technologies. Developed multiple agriculture innovation projects including an AI WhatsApp Farmer Assistant, Automatic Solar Irrigation System, and two MSME Idea Hackathon submissions. Seeking Graduate Engineer Trainee opportunities in Agricultural Machinery, AgriTech, Automation, and Product Development.

EDUCATION

Nandha Engineering College (Anna University) *2023 – 2027*

B.Tech in Agricultural Engineering, CGPA: 8.25 / 10 — Perundurai, Tamil Nadu

- Relevant Coursework: Farm Machinery, Soil & Water Engineering, Irrigation Engineering, Machine Design, Hydraulics, CAD, Precision Agriculture

Government Boys Higher Secondary School *2022 – 2023*

Higher Secondary Certificate (HSC), Percentage: 71% — Tamil Nadu

TECHNICAL SKILLS

- Design & Engineering: AutoCAD, SolidWorks, Engineering Graphics
- Automation & AI: No-Code Automation (n8n), OpenAI API, Prompt Engineering, IoT Basics
- Tools: Microsoft Excel, Microsoft PowerPoint, GitHub

CORE AGRICULTURAL SKILLS

- Farm Mechanization, Precision Agriculture, Hydroponics, Agricultural Processing, Post-Harvest Technology, Smart Farming, Food Processing, Sustainable Agriculture, Renewable Energy Systems

ACADEMIC & INNOVATION PROJECTS

Mechanized Tapioca (Kelangu) Sett Planting Machine *MSME Idea Hackathon 6.0*

- Designed a mechanized planter (concept design, mechanism layout, CAD modeling) to reduce manual labor and time in tapioca sett planting; submitted as an original engineering solution.

Farmer WhatsApp AI Chatbot (Mann Nanban) *In Progress*

- Built an AI-driven WhatsApp dispatcher (n8n, OpenAI API, Google Sheets, WhatsApp Business API) connecting farmers with service providers; designed the automation workflow and AI response logic end-to-end, handling real farmer queries in Tamil.

Automatic Solar Irrigation System *Academic Project*

- Designed a solar-powered irrigation system (Solar Panel, Soil Moisture Sensor, Relay Module) enabling automated pump operation to reduce unnecessary irrigation and water wastage.

Plant Residue Mat for Organic Mulching *Academic Project*

- Developed a biodegradable mulching mat from plant residue — a low-cost, eco-friendly alternative to plastic mulching for weed and moisture control.

INTERNSHIP / INDUSTRIAL TRAINING

Hykrops Hydroponic Industry — Intern *2025*

- Worked with hydroponic nutrient management and NFT systems; monitored EC & pH levels; studied irrigation scheduling and assisted in day-to-day cultivation operations.

Sakthi Sugar Mill — Intern *2025*

- Gained hands-on exposure to sugar manufacturing processes, plant operations, industrial safety practices, and quality control on the production floor.

INDUSTRIAL VISITS

- NABARD (agricultural finance & rural development), Milky Mist Dairy (dairy processing & quality control), Dharanee Agrovators (agricultural machinery & mechanization)

CERTIFICATIONS

- AI & ML in Farm Machinery — Workshop
- Innovation Day 2024 — Project Expo
- Inspire'25 — Paper Presentation

ACHIEVEMENTS

- MSME Idea Hackathon 6.0 — Submitted an original machine design project
- National Level Project Expo — Participant, Innovation Day 2024
- National Level Paper Presentation — Participant, Inspire'25
- Winner — Kabaddi, Annual Sports Meet 2024

LANGUAGES & SOFT SKILLS

- Languages: Tamil, English, Kannada
- Soft Skills: Problem Solving, Quick Learner, Team Collaboration, Leadership, Time Management, Communication, Adaptability